MARWAN EL SAABI GONZÁLEZ

DATA ANALYSIS & MACHINE LEARNING | MSC BIOINFORMATICS CANDIDATE
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PROFILE

Biologist turned data scientist, building ML systems end to end — from raw data to deployed tools: LLM-powered applications, supervised ML, time-series forecasting and relational data modelling. Seeking a data analyst or data science internship or junior role.

TECHNICAL PROJECTS & SOFTWARE DEVELOPMENT

AI-Driven Generative Application (De Novo Drug Designer) | *Python*, *Streamlit*, *Google Gemini API*, *RDKit*
- Architected and deployed an interactive open-source web application for *de novo* molecular generation and 3D rendering.
- Integrated the Gemini API with an RDKit validation layer to filter chemically invalid molecular structures.

Retail Demand Forecasting | *Python*, *LightGBM*, *Pandas*, *Streamlit*
- Forecast weekly SKU demand 4 weeks ahead from 1M+ transactions using LightGBM with lag, price and calendar features.
- Beat naive, seasonal-naive and moving-average baselines by 4.8% wMAPE under rolling-origin backtesting.

Alzheimer's Drug Classifier (ML Virtual Screening) | *Python*, *Scikit-learn*, *XGBoost*, *RDKit*
- Curated 9.7k ChEMBL IC50 records into 4.3k unique molecules, encoded as 1024-bit Morgan fingerprints.
- Benchmarked RF, SVM, XGBoost and KNN with scaffold-split CV: 0.92 AUROC (a random split inflates it to 0.96).

RNA-seq Analysis Dashboard | *Python*, *Streamlit*, *Plotly*, *Scikit-learn*
- Built an interactive Streamlit dashboard for RNA-seq analysis: PCA, volcano/MA plots and heatmaps.
- Implemented differential expression analysis (Welch's t-test, BH FDR correction) on a public dataset (Himes et al., 2014).

Melanoma Detection (Classical Computer Vision) | *Python*, *OpenCV*, *Scikit-learn*, *Scikit-image*
- Automatic segmentation of dermoscopic lesions via Otsu, K-means and watershed, selected by IoU against ground truth.
- Extracted 40 shape, color and texture (LBP/GLCM) features; SVM reached 90.2% accuracy and 84% melanoma sensitivity.

PROFESSIONAL EXPERIENCE

- **Delivery Driver (Part-time) / Teigal Logística (Amazon Delivery Partner)** | June 2026 to Present
- **Production Operator / Stellantis España S.L, Vigo** | Oct 2024 to Oct 2025
- **Seasonal Farmworker (Grape Harvest) / Iman Temporing ETT, S.L, for Lagar de Fornelos** | Sept 2024

EDUCATION

University of A Coruña (Universidade da Coruña), A Coruña | Sept 2025 to Feb 2027
MSc in Bioinformatics Applied to Health Sciences
University of Vigo (Universidade de Vigo), Vigo | Jul 2024
BSc in Biology (Specialization: Biotechnology)

AREAS OF EXPERTISE

- **Programming & Data:** Python (NumPy, Pandas, OpenCV), R (Tidyverse, ggplot2, DESeq2), SQL, Excel, MATLAB, Bash.
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, Neural Networks, Cross-Validation, Forecasting, LLM APIs (Gemini).
- **Tools & Infrastructure:** Linux, Git/GitHub, Docker, Conda, Nextflow, High-Performance Computing (SLURM).

LANGUAGES

Spanish and Galician (native), English (B2, CertAcles)

PUBLICATION

Alarcón-Ríos L., **Saabi González M.**, Velo-Antón G. (2026). Melanism in salamanders: effects of insularity, sex and size on dorsal colouration in *Salamandra salamandra*. *Journal of Zoology*. doi:10.1111/jzo.70133